

CLI Usage

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1. Using CLI for Prediction

CLI (Command-Line Interface) is a way for users to interact with computers or software programs by typing text commands to run programs or execute tasks, rather than clicking icons or buttons through a graphical user interface (GUI).

2. CLI Usage Syntax

```
yolo TASK MODE ARGS
```

where TASK (optional) is one of [detect, segment, classify, pose, obb]
MODE (required) is one of [train, val, predict, export, track, benchmark]
ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that override defaults.

3. Image Prediction

Use yolo26n.pt to predict images bundled with the ultralytics26 project: If the system cannot find the corresponding model file in the directory where the command is run, it will download automatically (if download fails, you can copy the model manually).

Navigate to the project folder:

```
cd /home/sunrise/.local/lib/python3.12/site-packages/ultralytics
```

Use yolo26n.pt to detect images in the target folder and output results:

```
yolo predict model=yolo26n.pt source='/home/sunrise/.local/lib/python3.12/site-packages/ultralytics/assets'
```

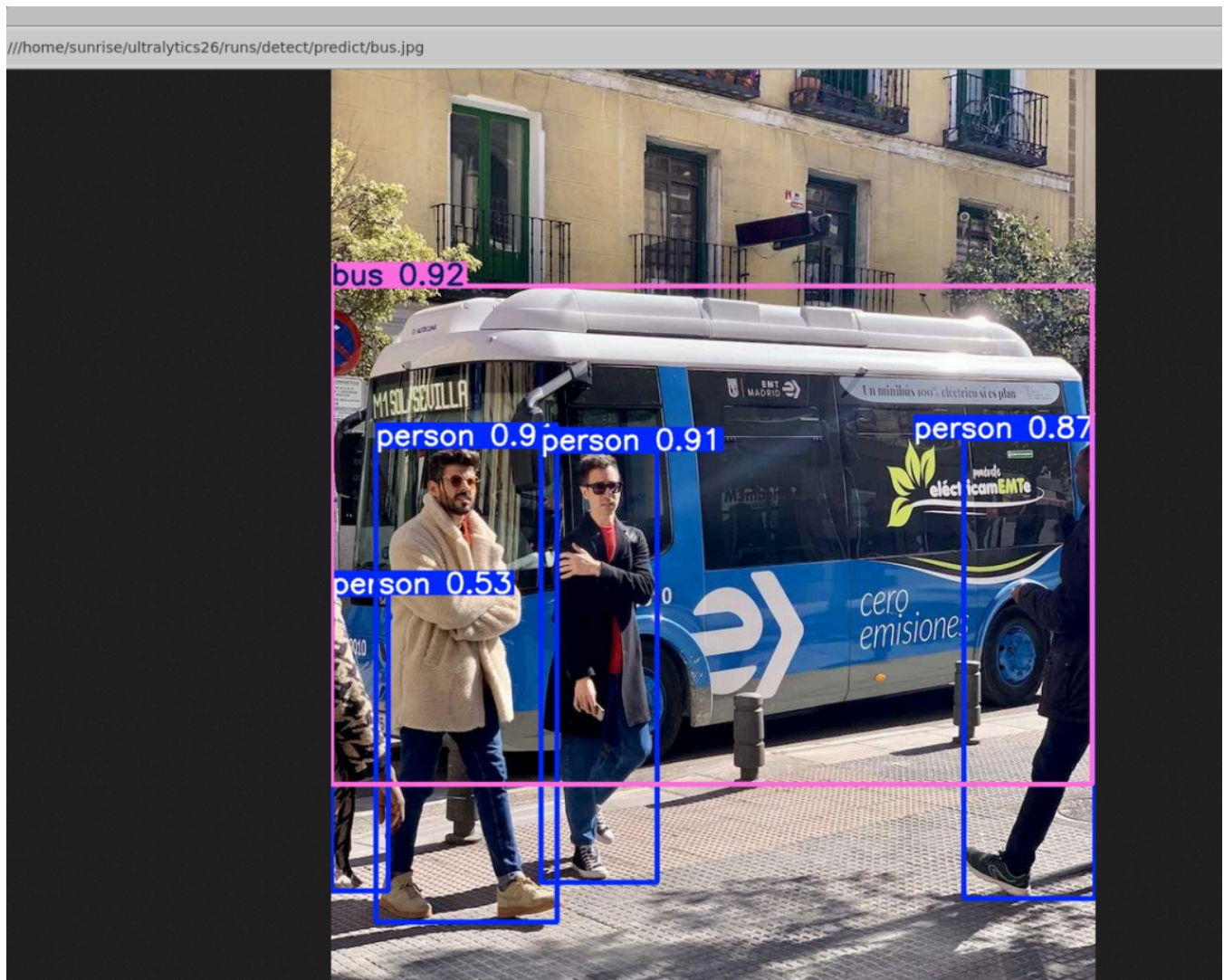
```
sunrise@ubuntu:~/local/lib/python3.12/site-packages/ultralytics$ yolo predict model=yolo26n.pt source='/home/sunrise/.local/lib/python3.12/site-packages/ultralytics/assets'
Creating new Ultralytics Settings v0.0.6 file 📄
View Ultralytics Settings with 'yolo settings' or at '/home/sunrise/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/quickstart/#ultralytics-settings.
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to 'yolo26n.pt': 100% 5.3MB 38.7KB/s 2:20
Ultralytics 8.4.66 Python-3.12.3 torch-2.12.0+cu130 CPU (Cortex-A78AE)
YOLO26n summary (fused): 122 layers, 2,408,932 parameters, 0 gradients, 5.4 GFLOPs

image 1/2 /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/assets/bus.jpg: 640x480 4 persons, 1 bus, 226.6ms
image 2/2 /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/assets/zidane.jpg: 384x640 2 persons, 1 tie, 187.6ms
Speed: 4.1ms preprocess, 207.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 640)
Results saved to /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/runs/detect/predict
? Learn more at https://docs.ultralytics.com/modes/predict
```

Preview

Output video location for yolo detection: /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/runs/detect/predict

Each run, the predict folder in this directory auto-increments by 1



4. Video Prediction

Use yolo26n.pt to predict videos from the ultralytics project (non-ultralytics bundled videos): If the system cannot find the corresponding model file in the directory where the command is run, it will download automatically (if download fails, you can copy the model manually).

Navigate to the project folder:

```
cd /home/sunrise/.local/lib/python3.12/site-packages/ultralytics
```

Use yolo26n.pt to detect videos in the target folder and output results:

```
yolo predict model=yolo26n.pt source='/home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos'
```

```

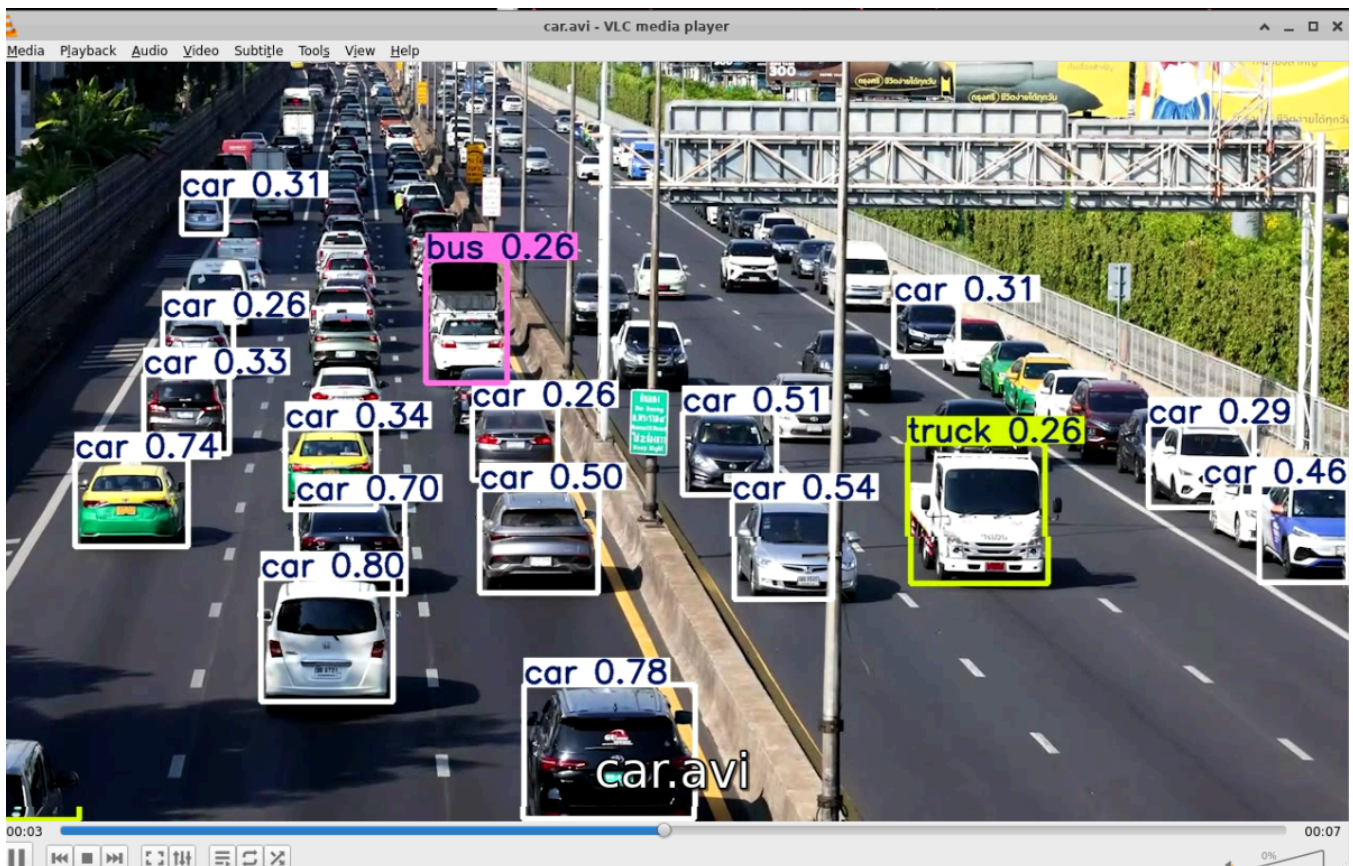
video 1/1 (frame 7/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 1 person, 10 cars, 169.5ms
video 1/1 (frame 8/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 1 person, 8 cars, 170.3ms
video 1/1 (frame 9/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 9 cars, 1 traffic light, 169.9ms
video 1/1 (frame 10/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.1ms
video 1/1 (frame 11/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.1ms
video 1/1 (frame 12/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.4ms
video 1/1 (frame 13/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 169.9ms
video 1/1 (frame 14/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.6ms
video 1/1 (frame 15/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.2ms
video 1/1 (frame 16/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 10 cars, 2 trucks, 170.3ms
video 1/1 (frame 17/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.1ms
video 1/1 (frame 18/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 169.7ms
video 1/1 (frame 19/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.8ms
video 1/1 (frame 20/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.3ms
video 1/1 (frame 21/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.2ms
video 1/1 (frame 22/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 170.7ms
video 1/1 (frame 23/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 2 trucks, 173.7ms
video 1/1 (frame 24/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 169.7ms
video 1/1 (frame 25/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 9 cars, 1 truck, 174.0ms
video 1/1 (frame 26/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 1 truck, 173.9ms
video 1/1 (frame 27/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 2 trucks, 173.3ms
video 1/1 (frame 28/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 8 cars, 2 trucks, 173.7ms
video 1/1 (frame 29/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 9 cars, 173.4ms
video 1/1 (frame 30/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 9 cars, 173.0ms
video 1/1 (frame 31/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 9 cars, 173.0ms
video 1/1 (frame 32/150) /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/videos/1986-153555214_medium.mp4: 384x640 9 cars, 173.0ms

```

Preview

Output video location for yolo detection: /home/sunrise/.local/lib/python3.12/site-packages/ultralytics/runs/detect/predict-2

Each run, the predict folder in this directory auto-increments by 1



5. Real-time Prediction

Use yolo26n.pt to predict USB camera feed: If the system cannot find the corresponding model file in the directory where the command is run, it will download automatically (if download fails, you can copy the model manually).

Navigate to the project folder:

```
cd /home/sunrise/.local/lib/python3.12/site-packages/ultralytics
```

Use yolo26n.pt to detect camera feed and output results: Currently, only USB cameras can directly use CLI for real-time prediction. There is no documentation found for CSI cameras directly as input sources.

```
yolo predict model=yolo26n.pt source=0 save=False show # Object Detection
# Instance Segmentation: yolo predict model=yolo26n-seg.pt source=0 save=False show
# Image Classification: yolo predict model=yolo26n-cls.pt source=0 save=False show
# Pose Estimation: yolo predict model=yolo26n-pose.pt source=0 save=False show
# Oriented Object Detection: yolo predict model=yolo26n-obb.pt source=0 save=False show
```

Press "Ctrl + C" in the terminal to terminate the program!

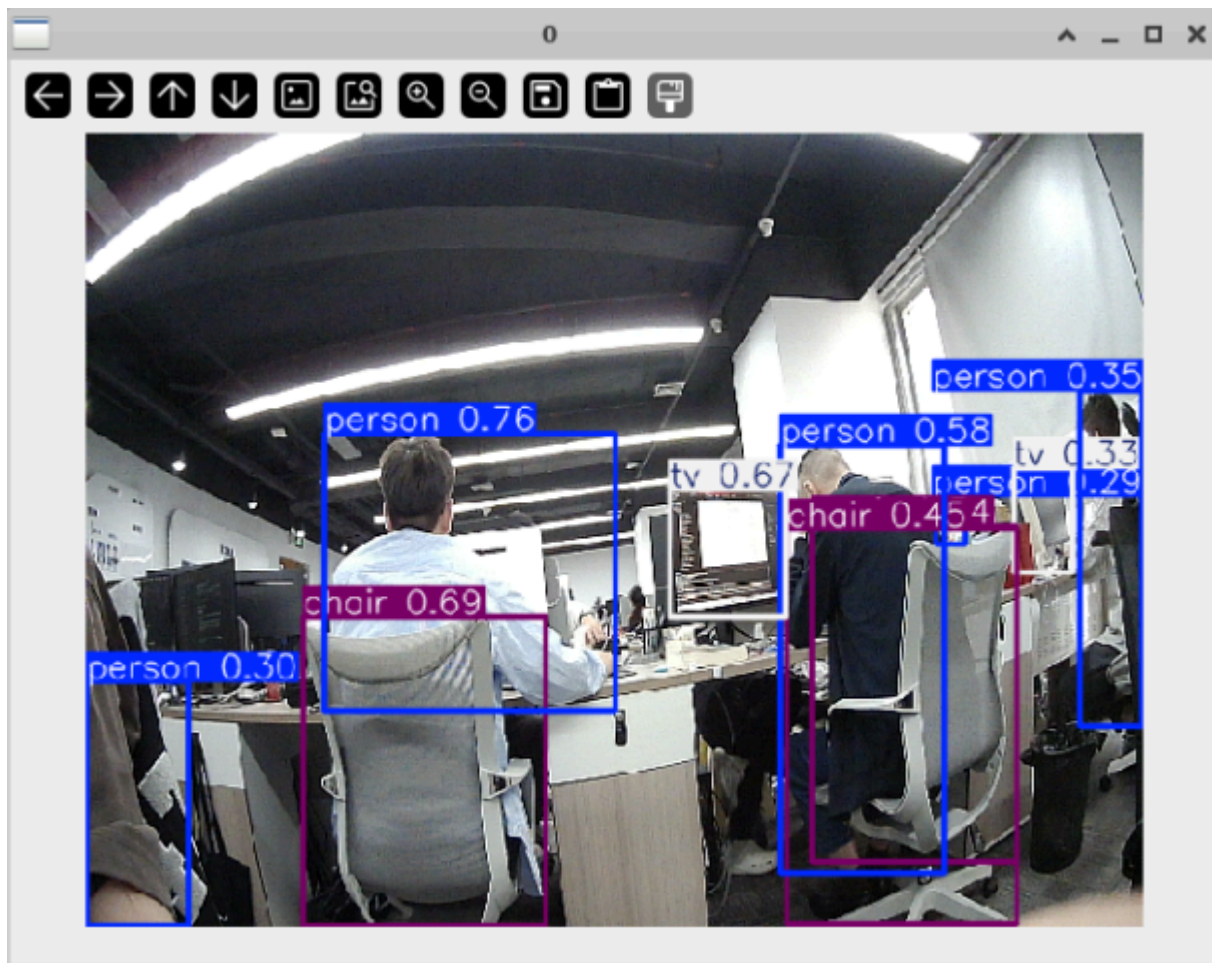
Parameter Description

`model`: Specify the YOLO model

`source`: Specify the detection source: Multiple cameras can be switched by numbers

`save=False`: Disable saving results

`show`: Real-time display



References

[Home - Ultralytics YOLO Documentation](#)

[Command-Line Interface - Ultralytics YOLO Documentation](#)